WORKED SOLUTION

STEP II 1987, Question 12

Mechanics • Equilibrium and Energy

STEP II 1987, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q12

The key idea

"Equilibrium in every position" is a statement about potential energy: if the bead can rest anywhere on the wire, the potential energy of the system can't change as the bead slides along it, so it must be constant on the wire. Writing that energy in terms of the bead's height and its distance from the ring gives the curve at once. The same fact makes the second part neat: with the potential energy constant, conservation of energy says the total kinetic energy of bead and particle is constant, and all that remains is some geometry linking the two speeds.

Solution

We model the bead and the hanging mass as particles, the string as light and inextensible, and the ring and wire as smooth. Put the ring at the origin O, with y measured vertically upwards. Let the bead be at polar coordinates (r, θ), with θ measured anticlockwise from the horizontal, so the bead is at height $y = r\sin\theta$ relative to O. Let L be the length of the string.

The equation of the curve

The string from O to the bead has length r, so the hanging particle is a distance $L - r$ below the ring. Taking O as the zero level, the potential energy of the system is

$$V = mgr\sin\theta - mg(L - r) = mg\,(r + r\sin\theta) - mgL.$$

The wire is smooth, so its reaction on the bead is perpendicular to it and the only forces that could move the system along the wire come from V. The system is in equilibrium at a position exactly when V is stationary there with respect to movement along the wire. Equilibrium at *every* position means $\mathrm{d}V/\mathrm{d}s = 0$ all along the wire (where s is arc length), so V is constant on the wire:

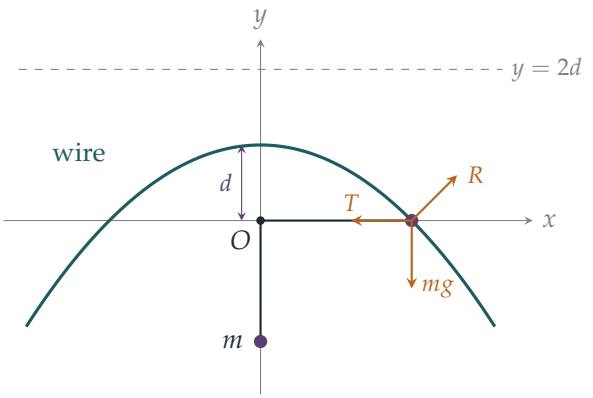
$$r(1 + \sin\theta) = c \quad \text{for some constant } c.$$

You can see this by resolving forces, too. In equilibrium the hanging particle gives tension mg throughout the string (the ring is smooth). On the bead, a tension mg towards O has tangential component $-mg\,\frac{dr}{ds}$, and the weight has tangential component $-mg\,\frac{dy}{ds}$. These must cancel, so $\frac{d}{ds}(r+y)=0$, which is the same condition.

To find c: since $1+\sin\theta \le 2$, we have $r = c/(1+\sin\theta) \ge c/2$, with equality when $\theta = \frac{\pi}{2}$ (the point of the wire directly above the ring). So the shortest distance from the ring to the wire is $c/2$, which gives $c = 2d$, and the curve is

$$\boxed{r(1+\sin\theta) = 2d} \qquad \text{i.e.} \qquad r = \frac{2d}{1+\sin\theta}.$$

(If you measure θ from the downward vertical instead, the same curve reads $r(1-\cos\theta) = 2d$.) Since $r = 2d - y$, every point of the wire is as far from O as it is from the horizontal line $y = 2d$. So the wire is a parabola with its focus at the ring and its directrix a height $2d$ above it; in Cartesian form, $x^2 = 4d(d-y)$, with vertex $(0, d)$.



In the picture the bead is level with the ring; the tension $T = mg$, the weight mg and the normal reaction R balance because the parabola's slope there is exactly -1.

The speed of the bead

Now the bead moves along the wire with speed u. The string stays taut and its length is fixed, so the hanging particle is still a distance $L - r$ below the ring.

? **Ask yourself** How fast is the hanging particle moving?

Its depth below the ring is $L - r$, so its velocity is vertical with speed $|\dot{r}|$. Note that $\dot{r}$ is only the radial part of the bead's velocity, not its whole speed u.

The tension is no longer mg once things move, but it does no net work on the system: the string is inextensible and the ring smooth, so the work it does on the bead is cancelled by the work it does on the hanging particle. The wire's reaction is perpendicular to the bead's motion and does no work. So energy is conserved:

$$\tfrac{1}{2}mu^2 + \tfrac{1}{2}m\dot{r}^2 + V = \text{constant}.$$

But on the wire $V = mg \cdot 2d - mgL$ is constant, by the first part. Hence

$$u^2 + \dot{r}^2 = \text{constant}. \qquad (*)$$

Next we link $\dot{r}$ to u. Differentiating $r = 2d/(1+\sin\theta)$,

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = -\frac{2d\cos\theta}{(1+\sin\theta)^2} = -\frac{r\cos\theta}{1+\sin\theta}, \qquad \text{so} \qquad \dot{r}^2 = r^2\dot{\theta}^2\,\frac{\cos^2\theta}{(1+\sin\theta)^2} = r^2\dot{\theta}^2\,\frac{1-\sin\theta}{1+\sin\theta},$$

using $\cos^2\theta = (1-\sin\theta)(1+\sin\theta)$. From the curve, $1+\sin\theta = 2d/r$ and $1-\sin\theta = 2-2d/r = 2(r-d)/r$, so

$$\frac{1-\sin\theta}{1+\sin\theta} = \frac{r-d}{d}, \qquad \text{giving} \qquad \dot{r}^2 = \frac{r-d}{d}\,r^2\dot{\theta}^2.$$

In polar coordinates the speed satisfies $u^2 = \dot{r}^2 + r^2\dot{\theta}^2$, so

$$u^2 = r^2\dot{\theta}^2\left(\frac{r-d}{d}+1\right) = \frac{r}{d}\,r^2\dot{\theta}^2, \qquad \text{and therefore} \qquad \dot{r}^2 = \frac{r-d}{r}\,u^2.$$

? Ask yourself What is $\dot{r}$ when $r = d$?

At $r = d$ the bead is at the vertex, $\theta = \frac{\pi}{2}$, where r takes its minimum value and $\mathrm{d}r/\mathrm{d}\theta = 0$. So $\dot{r} = 0$ there (as the formula $\dot{r}^2 = \frac{r-d}{r}u^2$ also says), and the hanging particle is momentarily at rest. The constant in $(*)$ is therefore v^2:

$$u^2 + \frac{r-d}{r}\,u^2 = v^2 \quad\Longrightarrow\quad u^2\,\frac{2r-d}{r} = v^2 \quad\Longrightarrow\quad u = \left(\frac{r}{2r-d}\right)^{\frac{1}{2}} v,$$

taking the positive root since u and v are speeds. This is the required result.

Checks

- At $r = d$ the formula gives $u = v$, as it must.
- As $r \to \infty$, $u \to v/\sqrt{2}$. Far out, the parabola's tangent points almost straight along the radius, so $\dot{r} \approx u$ and the kinetic energy $\frac{1}{2}mv^2$ is shared equally between bead and particle.
- Equilibrium at the level of the ring: the point $(2d, 0)$ is on the curve, where tension mg (horizontal) and weight mg (vertical) combine to a force at 45°, normal to the parabola's slope of -1 there.
- Numerically, with $d = 1.3$ and $v = 2$, we integrated the equation of motion of the bead-and-particle system along the parabola from the vertex. At $r = 2.8905$ the bead's speed was 1.606309, matching $\sqrt{r/(2r-d)}\,v$ to about 10^{-8}. We also confirmed that the tangential force on a bead at rest vanishes at 1000 random points of the wire.

Common slips

- Forgetting the hanging particle's kinetic energy, which gives $u = v$ for all r. The potential energy is constant, but the kinetic energy is shared.
- Taking the particle's speed to be u instead of $|\dot{r}|$. Only the radial part of the bead's motion pulls string through the ring.
- Measuring θ from a different line and then quoting $r(1 + \sin\theta) = 2d$. Any correct polar form is fine, but say which line θ is measured from.

Beyond STEP

The idea that a system is in equilibrium where its potential energy is stationary is the *principle of virtual work*, and it leads into Lagrangian mechanics. There, kinetic energy is written in terms of one generalised coordinate (here θ), and the equations of motion follow from it without ever finding the tension. Because V is constant on this wire, the bead and particle move as if no forces acted at all, and our relation is just conservation of the kinetic energy $\frac{1}{2}m\,r^2(2r - d)\dot{\theta}^2/d$. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers energy methods and constrained motion in this spirit.

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